

outlined: beyond the 32×/0.5× colour range ▽: slower than the column to its left corner: same configuration, auto-vectorizer on †: auto-vectorizer off in Vanilla Gibbon too

Real-world	ColorOctree	1×	1.09× 1.09×	▽ 0.721× 0.732×	0.815× 0.822×	1.16× 1.14×	1.17× 1.16×	1.17× 1.16×
	Compiler	1×	3.45× 3.47×	4.22× 4.4×	▽ 4.07× 4.15×	9.08× 9.25×	9.11× 9.35×	9.11× 9.35×
	DBQuery	1×	1.23× 1.23×	▽ 1.18× 1.16×	1.22× 1.22×	1.96× 1.96×	1.97× 1.96×	1.97× 1.96×
	DecisionTree	1×	1.83× 1.85×	2.77× 2.76×	2.77× 2.76×	2.77× 2.77×	2.77× 2.77×	2.77× 2.77×
	DecisionTreeClassify	1×	1.06× 1.05×	▽ 1.02× 0.985×	1.02× 0.984×	1.02× 0.983×	1.03× 0.969×	1.03× 0.969×
	DomTree	1×	1.29× 1.3×	1.33× 1.33×	▽ 1.26× 1.28×	1.35× 1.36×	▽ 1.34× 1.36×	1.34× 1.36×
	KDTree	1×	1.23× 1.23×	1.31× 1.21×	1.31× 1.21×	1.31× 1.21×	1.32× 1.21×	1.32× 1.21×
	ObjectGraph	1×	1.59× 1.61×	1.77× 1.59×	1.99× 2.05×	2.42× 2.42×	2.42× 2.41×	2.42× 2.41×
	OctTree	1×	1.04× 1.05×	▽ 0.935× 0.934×	1.01× 1.04×	1.42× 1.44×	1.42× 1.44×	1.42× 1.44×
	PiecewiseFunctions	1×	1.33× 1.34×	1.38× 1.39×	1.57× 1.59×	2.46× 2.45×	2.47× 2.45×	2.47× 2.45×
	Trie	1×	1.24× 1.25×	1.26× 1.27×	▽ 1.24× 1.25×	2.31× 2.31×	▽ 2.3× 2.28×	2.3× 2.28×
Synthetic	Add1TreeInt8	1×	1.16× 1.03×	1.18× 1.22×	1.75× 2.02×	2.05× 2.18×	2.1× 2.3×	2.1× 2.3×
	Add1TreeInt16	1×	1.3× 1.29×	▽ 1.21× 1.22×	1.8× 1.98×	2.08× 2.14×	2.18× 2.04×	2.18× 2.04×
	Add1TreeInt32	1×	1.14× 1.21×	1.19× 1.18×	1.62× 1.7×	1.89× 1.88×	▽ 1.88× 1.86×	1.88× 1.86×
	Add1TreeInt64	1×	1.14× 1.2×	1.14× 1.13×	1.46× 1.56×	1.62× 1.66×	1.62× 1.64×	1.62× 1.64×
	LinearListReduction	1×	2.23× 2.27×	▽ 1.99× 2.13×	2× 2.01×	3.74× 3.9×	3.98× 3.99×	3.98× 3.99×
	MonoTree	1×	1.63× 1.65×	1.65× 1.65×	2.05× 2.09×	2.22× 2.22×	2.24× 2.24×	2.24× 2.24×
	TernaryTree	1×	1.12× 1.14×	1.12× 1.1×	1.32× 1.29×	1.37× 1.35×	▽ 1.33× 1.36×	1.33× 1.36×
Arithmetic intensity	ArithmeticIntensityInt8 †	1× 1.18×	▽ 0.963× 1.22×	1.03× 1.26×	1.1× 3.32×	1.14× 3.56×	3.67× 3.56×	3.67× 3.56×
	ArithmeticIntensityInt16 †	1× 1.99×	1.04× 2.17×	▽ 1.03× 2.24×	1.08× 3.69×	1.12× 4.11×	3.85× 3.9×	3.85× 3.9×
	ArithmeticIntensityInt32 †	1× 1.31×	1.03× 1.37×	▽ 1.02× 1.43×	1.08× 2.36×	1.12× 2.43×	2.55× 2.53×	2.55× 2.53×
	ArithmeticIntensityInt64 †	1× 0.968×	▽ 0.997× 1.05×	1.03× 1.03×	1.04× 1.05×	1.08× 1.08×	1.08× 1.08×	1.08× 1.08×
		Vanilla Gibbon (AoS)	Mutable cursors (AoS)	SoA layout	Loopification	Buffer sharing	Gibbon SIMD	Total